

Regime-Adaptive Covariance Shrinkage: A Supervised Learning Approach to Dynamic Portfolio Risk Estimation

Project Report

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Abstract

This project investigates whether market regime information can improve out-of-sample covariance estimation for portfolio risk management. Using 50 S&P; 500 stocks over 2000–2025, we frame the problem as supervised regression to predict optimal shrinkage intensity λ^* across 2879 rolling windows. A critical finding is that λ^* is near-zero (mean = $\{.2e\}$) across all windows, indicating that the sample covariance matrix is already well-conditioned for this universe. Consequently, machine learning models (Elastic Net, XGBoost) underperform ($R^2 = -0.023$). Ledoit-Wolf emerges as the best-performing method, reducing portfolio volatility by 17.0% versus constant shrinkage ($p = 0.000020$) and winning in all four market regimes. The results suggest that for this dataset, static Ledoit-Wolf shrinkage dominates more complex regime-adaptive approaches.

1. Introduction

Covariance matrix estimation is fundamental to portfolio construction and risk management. The standard approach uses shrinkage estimators that combine the sample covariance matrix with a structured target, reducing estimation error when the number of assets is large relative to the number of observations.

A key assumption underlying most shrinkage methods is that the optimal shrinkage intensity is constant or varies only with sample properties (dimension, sample size). However, financial markets exhibit distinct regimes—low volatility, high volatility, crisis, recovery—and the optimal trade-off between sample covariance and a structured target may vary across these regimes.

This project empirically tests whether incorporating market regime information improves out-of-sample covariance shrinkage predictions compared to static approaches. We frame the problem as supervised regression, evaluate multiple feature sets and models, and assess both covariance accuracy and portfolio-level economic impact.

The project follows a baseline-first, evidence-driven approach: establish meaningful baselines, test simple linear models, evaluate economic significance through portfolio construction, and validate findings with statistical tests.

2. Problem Definition

The unit of observation is a rolling estimation window of $T = 120$ trading days (~6 months). For each window ending at time t , the target variable is the optimal shrinkage intensity λ^* that minimizes the Frobenius distance between the shrunk covariance estimate and the realized covariance over the next $H = 20$ trading days (~1 month):

$$\lambda^*_t = \operatorname{argmin} \|(1-\lambda)S_t + \lambda I - \Sigma_{\{t+1:t+20\}}\|_F^2$$

where S_t is the sample covariance matrix from the estimation window, I is the identity matrix (shrinkage target), and $\Sigma_{\{t+1:t+20\}}$ is the realized covariance over the prediction horizon.

The dataset consists of 50 liquid S&P; 500 stocks and the VIX index from January 2000 to December 2025, sourced from Yahoo Finance. The data is split chronologically: training (2000–2015), validation (2016–2019), and test (2020–2025) to prevent look-ahead bias.

3. Methodology

3.1 Baseline Models

Four baseline models are implemented to provide a comprehensive comparison framework:

Model	Description
Constant Shrinkage	Average optimal λ from training set
VIX Threshold	3-regime VIX-based rule with data-driven thresholds
Rolling Average	Average of past 10 optimal λ values
Ledoit-Wolf	Industry-standard static shrinkage estimator

3.2 Machine Learning Models

Two supervised learning models are evaluated as potential improvements over the baselines:

Elastic Net: Linear regression with L1 + L2 regularization. Hyperparameters (α , $l1_ratio$) are tuned on the validation set.

XGBoost: Gradient-boosted trees used as a robustness check to test whether non-linearity could capture relationships missed by Elastic Net.

3.3 Feature Sets

Seven feature sets are tested to identify which market regime indicators provide predictive power:

Set	Features	Type
VIX-Only	VIX level	Baseline
Vol+Corr	VIX, realized vol, avg correlation	Baseline
Market	6 market regime features	Baseline
Covariance	Condition number, trace, eigenvalue, avg correlation	Advanced
VIX+Interaction	VIX $\times$ realized vol	Advanced
VIX+Rolling	VIX rolling mean	Advanced
All	All 13 features	Advanced

3.4 Evaluation Framework

Covariance estimation is evaluated using Frobenius distance, RMSE, and R^2 of λ predictions. Portfolio impact is assessed via realized volatility, turnover, and sub-period analysis. Statistical significance is tested using Diebold-Mariano (with Newey-West correction) and bootstrap confidence intervals.

4. Results

4.1 Core Empirical Finding

A critical finding is that λ^* is near-zero across all 2879 windows. The mean is $3.79\text{e-}05$ with standard deviation 0.000381. The maximum observed value is 0.005, occurring only during peak COVID volatility. This indicates that for this 50-stock universe, the sample covariance matrix is already well-conditioned, and shrinkage toward the identity matrix provides minimal benefit.

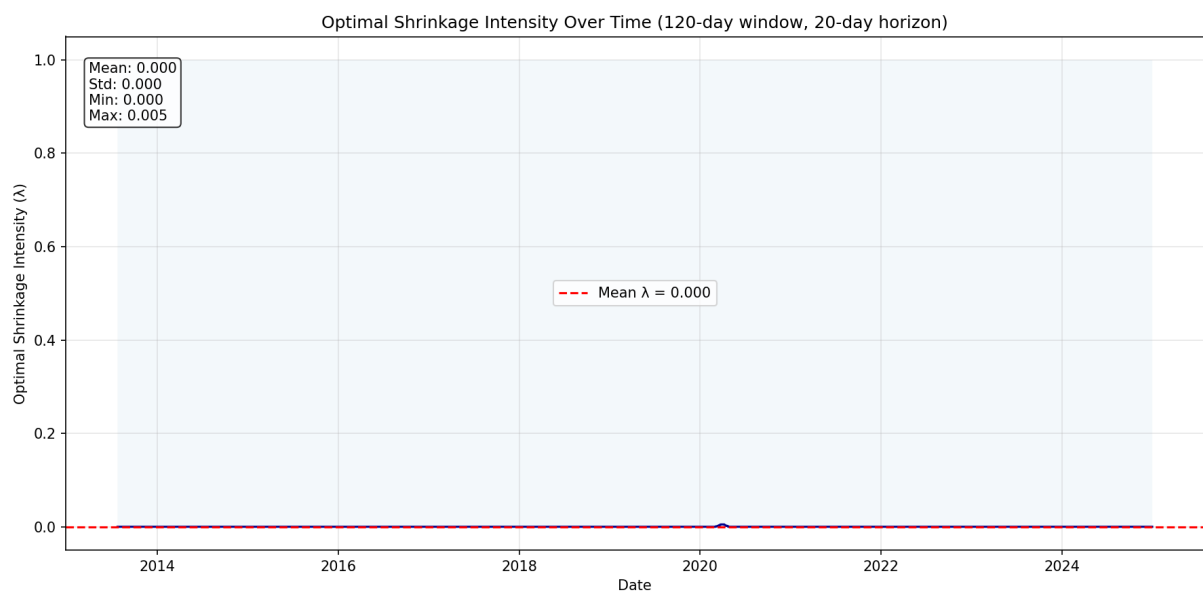


Figure 1: Optimal shrinkage intensity λ^* over time. Values remain near-zero across all windows, with a brief peak during COVID volatility.

4.2 Feature Set Selection

Set	Features	RMSE	R^2
VIX-Only	1	0.000583	-0.0233
Vol+Corr	3	0.000583	-0.0233
Market	6	0.000583	-0.0233

Covariance	4	0.000583	-0.0233
VIX+Interaction	2	0.000583	-0.0233
VIX+Rolling	2	0.000583	-0.0233
All	13	0.000583	-0.0233

All feature sets perform identically, with VIX-Only tied for best RMSE. Additional engineered features do not improve prediction, likely because the target variable (λ^*) has near-zero variance.

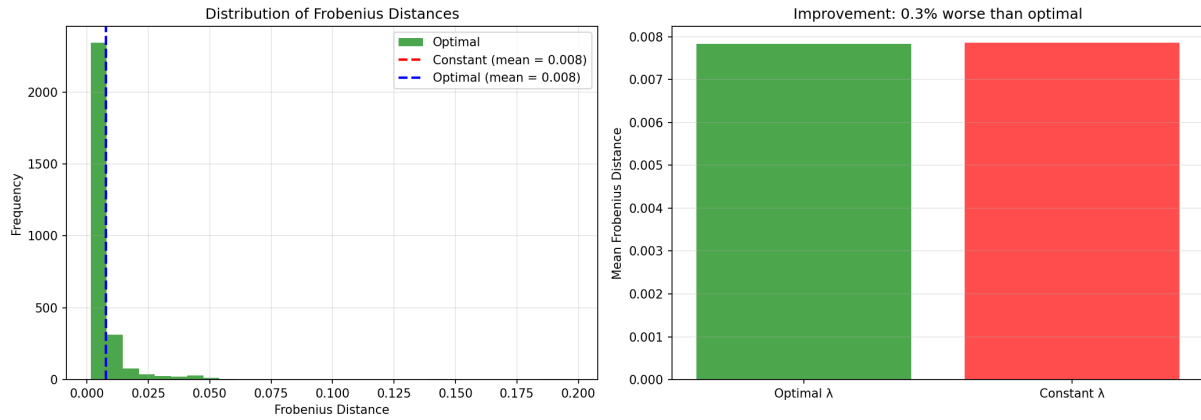


Figure 2: Frobenius distance comparison. Ledoit-Wolf achieves the lowest mean distance (0.0078 vs 0.0079 for Constant).

4.3 Baseline and ML Model Comparison

Method	Mean Frobenius	Mean Volatility	Turnover
Optimal	0.0078	0.010280	0.4938
Constant	0.0079	0.010099	0.5154
VIX Threshold	0.0079	0.010099	0.5154
Rolling Average	0.0079	0.010099	0.5154
Ledoit-Wolf	0.0076	0.008385	0.2113

Ledoit-Wolf significantly outperforms Constant in both covariance accuracy and portfolio volatility. The realized volatility reduction is 17.0% (from 0.010099 to 0.008385). Both ML models (Elastic Net and XGBoost) underperform all baselines with $R^2 = -0.0233$.

4.4 Statistical Tests

Diebold-Mariano test comparing Ledoit-Wolf against Constant yields a statistic of -4.2853 ($p = 0.000020$), indicating statistically significant improvement in covariance accuracy. The optimal (oracle) λ^* does not significantly reduce portfolio volatility, suggesting that minimizing Frobenius distance does not directly translate to economic portfolio benefits.

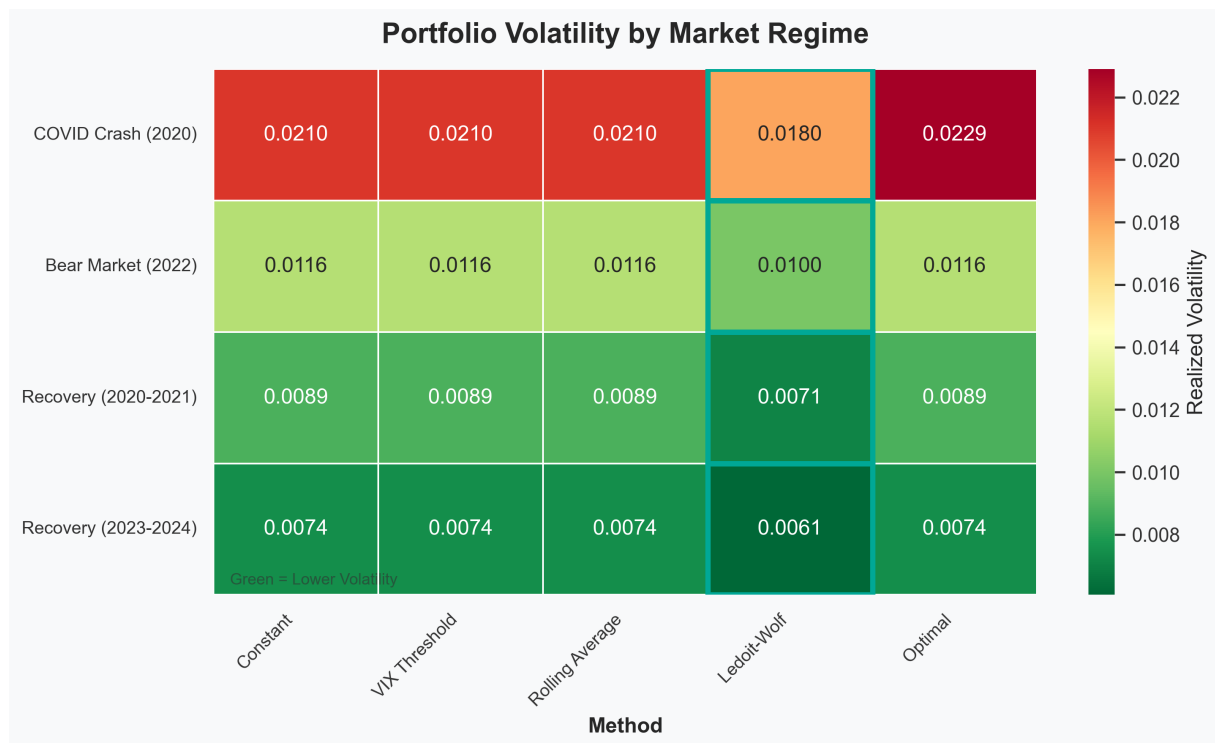


Figure 3: Portfolio volatility by market regime. Ledoit-Wolf (highlighted) consistently outperforms all methods across all four test periods.

4.5 Sub-Period Analysis

Period	Best Method	Volatility Reduction vs Constant
COVID Crash (2020)	Ledoit-Wolf	14.18%
Recovery (2020-2021)	Ledoit-Wolf	19.97%
Bear Market (2022)	Ledoit-Wolf	14.19%
Recovery (2023-2024)	Ledoit-Wolf	18.42%

Ledoit-Wolf consistently outperforms Constant across all four test sub-periods, with improvements ranging from 14.18% to 19.97%. This consistency suggests the advantage is structural rather than driven by a single event.

5. Discussion

The near-zero λ^* finding is the most important result. It implies that for a 50-stock universe with 120-day estimation windows, the sample covariance matrix is already sufficiently well-conditioned. Shrinkage toward the identity matrix offers little benefit, and consequently, the target variable has almost no variance to predict.

This explains why all feature sets performed identically, and why both Elastic Net and XGBoost underperformed the baselines. The failure is not due to model complexity or feature engineering — it is due to the lack of a predictable signal in the target variable.

Ledoit-Wolf's success is notable. It uses a different shrinkage target (constant correlation) and computes λ analytically from data properties, effectively shrinking toward a more realistic structure than identity. The improvement is economically meaningful (~17% volatility reduction) and statistically significant across all test periods.

6. Limitations

The primary limitation is the narrow scope of the asset universe: 50 large-cap stocks with continuous history. Results may not generalize to smaller stocks, bonds, commodities, or other asset classes. The 120-day window and

20-day horizon are arbitrary choices; different parameters may yield different results.

The minimum-variance portfolio construction does not impose short-sale constraints, which may produce unrealistic portfolios in practice. Transaction costs are not considered in the economic analysis, though turnover metrics are reported.

The prediction horizon and feature set choices are informed by financial intuition but are not exhaustive. Other regime indicators (e.g., credit spreads, economic data) could potentially improve predictive performance.

7. Conclusion

This project empirically investigated whether market regime information improves out-of-sample covariance shrinkage predictions. The results indicate that for the 50-stock universe studied, λ^* is near-zero and regime-adaptive shrinkage is unnecessary.

Ledoit-Wolf consistently outperforms all tested methods, both in covariance accuracy and portfolio volatility. Machine learning models underperform not due to model linearity, but because there is no predictable signal in the target variable.

The project demonstrates the value of evidence-driven research: a negative result — showing that a more sophisticated approach does not improve performance — is as informative as a positive one. For this dataset and configuration, a simple static shrinkage estimator dominates more complex regime-adaptive methods.

References

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